$2^3 = 8$ $\quad \log_2 8 = 3$ $\qquad 16 = 2^x$ $\quad \log_2 16 = x$

Week 6 - Part 2 - Osprey1st College Algebra - Summer 2024

Section covered: 9.3, 9.4, and 9.5

The horizontal line test shows that exponential functions are one-to-one and thus have inverse functions. The equation defining the inverse of a function is found by interchanging x and y in the equation that defines the function. Starting with $y = a^x$ and interchanging x and y yields $x = a^y$.

Logarithm
For all real numbers y and all positive numbers a and x, where $a \neq 1$,

$$y = \log_a x \quad \text{is equivalent to} \quad x = a^y.$$

The expression $\log_a x$ represents the exponent to which the base a must be raised in order to obtain x.

Problem 12. * Evaluate the given logarithms.

(a) $\log_5 25 = x$ $\quad 5^x = 25$ $\quad x = 2$

(b) $\log_2 8 = x$ $\quad 2^x = 8$ $\quad x = 3$

(c) $\log_3 1 = x$ $\quad 3^x = 1$ $\quad x = 0$

(d) $\log_4 \frac{1}{4} = x$ $\quad 4^x = \frac{1}{4}$ $\quad x = -1$

(e) $\log_3 \frac{1}{9} = x$ $\quad 3^x = \frac{1}{9}$ $\quad x = -2$

Problem 13. * Rewrite each equation as requested.

Logarithmic Equation	**Exponential Equation**
$\log_2 32 = 5$	$2^5 = 32$
$\log_6 \frac{1}{36} = -2$	$6^{-2} = \frac{1}{36}$
$\log_5 \frac{1}{125} = -3$	$5^{-3} = \frac{1}{125}$
$\log_{12} 12 = 1$	$12^1 = 12$
$\log_6 1 = 0$	$6^0 = 1$

Common Logarithms

Two of the most most important bases for logarithms are 10 and e. Base 10 logarithms are called common logarithms. The common logarithm of x is written $\log x$ where the base is understood to be 10.

$\log_{10} 100 = x \quad 10^x = 100 \quad x = 2$

Common Logarithm
For all positive numbers x,

$$\log x = \log_{10} x.$$

Natural Logarithm

The number e is defined as the value that $\left(1+\frac{1}{n}\right)$ approaches as n gets larger and larger, that is

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \approx 2.718281828459$$

The irrational number e, approximately 2.72, is called the **natural base**.

Logarithms with base $\boldsymbol{e}$ are natural logarithms because they occur in the life sciences and economics in natural situations that involve growth and decay. The base $\boldsymbol{e}$ logarithm of x is written $\mathbf{ln}\,\boldsymbol{x}$ (read **"el-en x"**). The expression $\ln x$ represents the exponent to which e must be raised in order to obtain x.

Natural Logarithm
For all positive numbers x,

$$\ln x = \log_e x.$$

Problem 14. Evaluate the given logarithms.

(a) $\log 1000 = x \quad 10^x = 1000 \quad x = 3$

(b) $\log \frac{1}{100} = x \quad 10^x = \frac{1}{100} \quad x = -2$

(c) $\ln 1 = x \quad e^x = 1 \quad x = 0$

(d) $\ln e = x \quad e^x = e \quad x = 1$

Problem 15. * Rewrite each equation as requested.

(a) Rewrite as an exponential equation.

$e^7 = x \qquad \ln x = 7$

(b) Rewrite as a logarithmic equation.

$\ln 2 = y \qquad e^y = 2$

(c) Rewrite as an exponential equation.

$e^y = 4 \qquad \ln 4 = y$

(d) Rewrite as a logarithmic equation.

$\ln x = 6 \qquad e^6 = x$

Logarithmic Functions

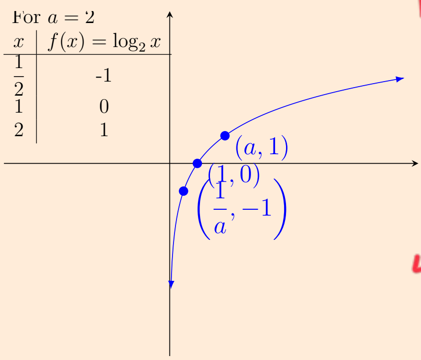
Logarithmic Function
If $a > 0$, $a \neq 1$, and $x > 0$, then the **logarithmic function with base a** is

$$f(x) = \log_a x.$$

Characteristics of the Graph of $f(x) = \log_a x$

1. The point $\left(\frac{1}{a}, -1\right)$, $(1, 0)$ and $(a, 1)$ are on the graph.
2. If $a > 1$, then f is and increasing function. If $0 < a < 1$, then f is a decreasing function.
3. The y-axis is a vertical asymptote.
4. The domain is $(0, \infty)$ and the range is $(-\infty, \infty)$.

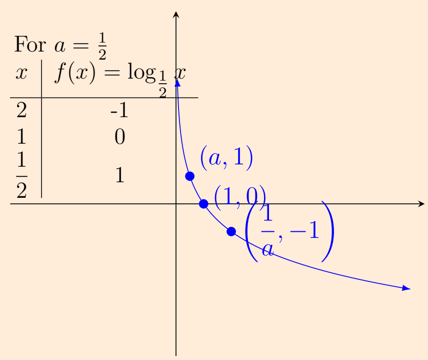
$f(x) = \log_a x$, $a > 1$



$\log_2 x$

x	y
$\frac{1}{2}$	-1
1	0
4	2

$f(x) = \log_a x$, $0 < a < 1$



Properties of Logarithms
For $x > 0, y > 0, a > 0, a \neq 1$, and any real number r, the following properties hold.

Product Rule	$\log_a xy = \log_a x + \log_a y$
Quotient Rule	$\log_a \frac{x}{y} = \log_a x - \log_a y$
Power Rule	$\log_a x^r = r \log_a x$
Logarithm of 1	$\log_a 1 = 0$
Base a Logarithm of a	$\log_a a = 1$

$x^3 \cdot x^4 = x^7$

$\frac{x^5}{x^3} = x^2$

Note: **There is no property of logarithms to rewrite a logarithm of a sum or difference.**

Problem 16. * Fill in the missing values to make the equations true.

(a) $\log_8 5 - \log_8 11 = \log_8 \square$ $\frac{5}{11}$

(b) $\log_9 \square + \log_9 12 = \log_9 60$ 5

(c) $\log_6 25 = \square \log_6 5$ 2 5^2

Problem 17. * Rewrite each expression. Assume all variables represent positive real numbers, with $a \neq 1$ and $b \neq 1$.

(a) $\log_7(8 \cdot 6)$ $= \log_7 8 + \log_7 6$

(b) $\log_6 \frac{5}{9}$ $= \log_6 5 - \log_6 9$

(c) $\log_a \sqrt[3]{m^2}$ $= \log_a m^{\frac{2}{3}} = \frac{2}{3}\log_a m$

(d) $\log\left(\frac{\sqrt[3]{x^7 z}}{y^2}\right)$ $= \log\left(\frac{x^{\frac{7}{3}} z^{\frac{1}{3}}}{y^2}\right) = \frac{7}{3}\log x + \frac{1}{3}\log z - 2\log y$

(e) $\log_a \frac{x^5}{z^4 \sqrt[5]{y^3}}$ $= \log\left(\frac{x^5}{z^4 y^{\frac{3}{5}}}\right) = 5\log x$

(f) $\log \sqrt[3]{\frac{xy^4}{z^2}}$

(g) $\log_4(x^2 + y^2)$

Problem 18. * Write each expression as a single logarithm with coefficient 1. Assume all variables represent positive real numbers, with $a \neq 1$ and $b \neq 1$.

(a) $\log_4 x - \log_4 y + \log_4 z$

(b) $\frac{1}{3}\log_a x + \frac{2}{3}\log_a y - \log_a xy$

(c) $6\log_m(3z+1) + \dfrac{1}{4}\log_m(z+7)$

(d) $4(\log_c y - 4\log_c w) + 3\log_c x$

(e) $3\log_m x + 2(\log_m w - 2\log_m y)$

Problem 19. * Use the properties of logarithms to evaluate each of the following expressions.

(a) $\log_2 36 - 2\log_2 3 =$

(b) $\ln e^{-8} + \ln e^{3} =$

(c) $2\ln_5 3 - \log_5 45 =$

Logarithms with Other Bases

We can use a calculator to find the values of either natural logarithms (base e) or common logarithms (base 10). However, sometimes we must use logarithms with other bases. The change-of-base theorem can be used to convert logarithms from one base to another.

Change-of-Base Theorem
For any positive real number x, a and b, where $a \neq 1$ and $b \neq 1$, the following holds.

$$\boldsymbol{\log_a x = \frac{\log_b x}{\log_b a}.}$$

In particular,

$$\boldsymbol{\log_a x = \frac{\log x}{\log a} \quad \text{or} \quad \log_a x = \frac{\ln x}{\ln a}.}$$

Problem 20. Use the change-of-base theorem to write each logarithm (i) in terms of common logarithms, (ii) in terms of natural logarithms. Approximation the answer to four decimal places for each logarithm.

(a) $\log_4 20$

(b) $\log_2 0.7$

Solving an Exponential or Logarithmic Equation
To solve an exponential or logarithmic equation, change the given equation into one of the following forms, where a and b are real numbers, $a > 0$and $a \neq 1$, and follow the guidelines.

1. $a^{f(x)} = a^{g(x)}$
The given equation is equivalent to the equation $f(x) = g(x)$. Solve algebraically.

2. $a^{f(x)} = b$
Solve by taking natural (or common) logarithms on each side $\ln a^{f(x)} = \ln b$ and then bring $f(x)$ in front of the logarithm, $f(x) \ln a = \ln b$.

3. $\log_a f(x) = b$
Solve by changing to exponential form $a^b = f(x)$.Check that each proposed solution is in the domain.

4. $log_a f(x) = log_a g(x)$
The given equation is equivalent to the equation $f(x) = g(x)$. Solve algebraically. Check that each proposed solution is in the domain.

Exponential equations

Problem 1. * Solve each exponential equation

(a) $64 = 32^{-x+2}$

(b) $10^{-8x} = 100^{2-3x}$

(c) $36^{3x-5} = 6^{8x}$

Problem 2. * Solve each equation. Give exact values using either base-10 or base-e logarithms. Then round your answer to the nearest hundredth. Do not round any intermediate computations.

(a) $5^{-6x} = 15$

(b) $2^{x+8} = 3$

(c) $2^{-x+10} = 5^{-6x}$

(d) $13^{-x+10} = 4^{2x}$